

AutoML_Experiment_R...
PDF

You are a doctor. You provided me with several audio files to help build an AI model for identifying respiratory diseases. I have now trained the model and generated a report. Please let me know whether the report is helpful to you.

I've carefully reviewed your report: 📄 AutoML_Experiment_Report__medic...

From a **clinician's perspective (respiratory medicine / screening use-case)**, here is a precise and practical evaluation of whether this report is actually helpful:

1. Overall judgment

Yes — the report is helpful, but mainly as a *technical validation document*, not yet a *clinical decision-support report*.

It demonstrates:

- Solid modeling rigor
- Reasonable performance
- Thoughtful uncertainty handling

However, it still lacks **direct clinical interpretability and actionability at the patient level**.

2. What is genuinely useful for me as a doctor

(1) Multi-modal design aligns with clinical reasoning

- Combining **symptoms (tabular)** + **audio (cough, vowel)** is clinically sound
- The finding that audio features are *complementary* (low correlation with tabular) is important
 - This suggests the model is not just re-learning questionnaire data

📄 AutoML_Experiment_Report__medic...

✓ Clinically meaningful

(2) Performance is acceptable for screening (not diagnosis)

- Macro-F1 = **0.8384**
- Accuracy $\approx$ **85%**  AutoML_Experiment_Report_medec...

Interpretation:

- This is **good for triage / screening**
- Not sufficient for **definitive diagnosis**

✓ I would consider this for:

- Pre-screening
- Risk stratification
- Remote monitoring

(3) Uncertainty analysis is the strongest clinical contribution

This is the most valuable part.

You show:

- Wrong predictions $\rightarrow$ higher uncertainty
- Clear separability between correct vs incorrect predictions

 AutoML_Experiment_Report_medec...

Clinical implication:

- Enables **human-in-the-loop system**
- Allows:
 - Low uncertainty $\rightarrow$ auto decision
 - High uncertainty $\rightarrow$ refer to clinician

✓ This is **deployable logic**, not just theory

(4) Two-tier workflow proposal is clinically realistic

Your recommendation:

- Automated triage
- Expert review for high-risk / high-uncertainty cases

 AutoML_Experiment_Report_medec...

✓ This matches real workflows in:

- Telemedicine
- Community screening

- Overloaded ED systems
-

3. What is missing (critical from a doctor's perspective)

(1) No clinical meaning of "Class 0/1/2"

This is a major issue.

You report:

- "Class 0, 1, 2"

But as a clinician I need:

- What diseases?
 - Asthma?
 - COPD?
 - Infection?
- Or severity levels?

! Without this → I cannot interpret predictions clinically

(2) No patient-level explanation

You do NOT provide:

- "Why was this patient classified as Class 2?"
- Feature attribution (e.g., SHAP, attention)

Even though you mention correlations:

- age
- wheezing
- smoking

These are **population-level**, not **individual-level explanations**

! This limits trust

(3) No clinical validation metrics

You report:

- Macro-F1

- Accuracy

But missing:

- Sensitivity (recall per disease)
- Specificity
- PPV / NPV

These are what clinicians actually use.

(4) Dataset is small and unclear clinically

- Only **546 patients**  AutoML_Experiment_Report__medic...
- No info on:
 - Clinical diagnosis standard (gold standard?)
 - Hospital vs community data
 - Device variability

! Generalizability is uncertain

(5) No deployment scenario validation

You propose workflow, but no:

- Prospective validation
 - Real-world test
 - Comparison vs clinicians
-

4. What I would request before using this clinically

To make this **clinically useful**, I would require:

A. Clinical labeling clarity

- Define each class clinically
 - Map to ICD or diagnostic categories
-

B. Patient-level explanation

For each prediction:

- Key contributing factors:
 - e.g., "wheezing + cough spectral pattern → asthma-like"

C. Clinical metrics

Report:

- Sensitivity / specificity per class
 - Especially:
 - False negatives (critical!)
-

D. External validation

- Different hospitals / devices
 - Different populations
-

E. Example cases

Show:

- 5–10 real patient cases
 - Input → prediction → explanation
-

5. Bottom-line answer (as your “doctor”)

Is the report helpful?

👉 Yes, but at the “model development” level, not yet at the “clinical decision” level.

- ✔ Strong for AI researchers / system validation
 - ✔ Promising for screening tool
 - ! Not yet sufficient for clinical adoption
-

6. If you want a stronger doctor-facing version

I can help you rewrite this report into a **clinician-oriented version**, including:

- Clinical interpretation of outputs
- Case-based explanations
- Deployment protocol (ED / telehealth)
- Regulatory-style validation framing

Just tell me 👉